Matching Intro

STSCI/INFO/ILRST 3900: Causal Inference

27 Oct 2026

Learning goals for today

At the end of class, you will be able to:

1. Explain how matching can be used to estimate causal effects
2. Explain bias variance trade-off in various matching procedures

Logistics

- Quiz 3: Nov 5
- Quiz 4: Dec 2 (Wednesday)
- Project Part 1 due Nov 6 (11:59pm)
- Full Report due Dec 7 (11:59pm)

Causal effect

What is the causal effect of a job training program on income?

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- **Average Treatment Effect** (on everyone)

$$\mathrm{E}(Y^{a=1}) - \mathrm{E}(Y^{a=0})$$

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- **Average Treatment Effect on the Treated** (ATT)

$$\mathrm{E}(Y^{a=1} \mid A = 1) - \mathrm{E}(Y^{a=0} \mid A = 1)$$

Matching: The big idea

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Goal: $\mathsf{E}(Y^{a=1} \mid A = 1) - \mathsf{E}(Y^{a=0} \mid A = 1)$ **ATT**

$$\mathsf{E}(Y^{a=1} \mid A = 1) \approx \frac{1}{n_t} \sum_{i:A_i=1} Y_i^{a=1} = \frac{1}{n_t} \sum_{i:A_i=1} Y_i$$

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Potential Solution:

For each treated individual i, get a set of **untreated** individuals, $\mathcal{M}_i$, and use their observations to estimate $Y_i^{a=0}$

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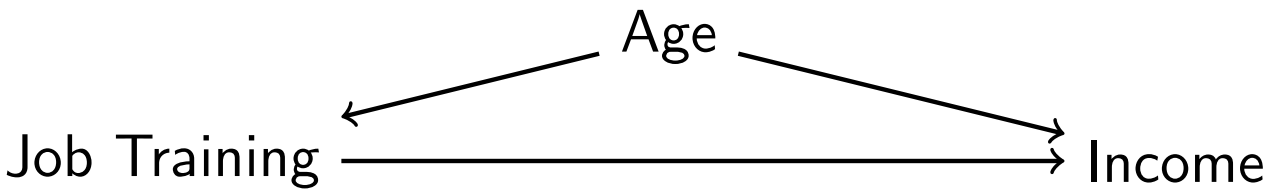
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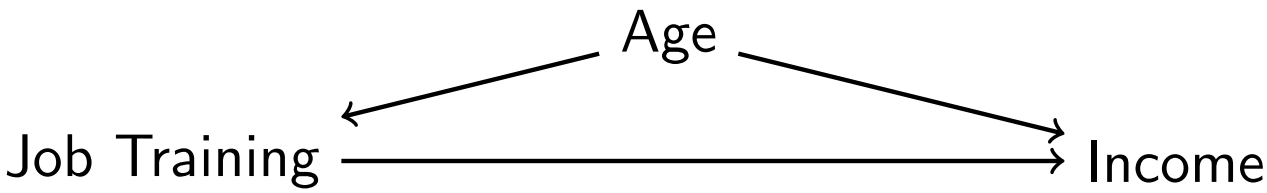
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$$\mathrm{E}(Y^{a=0} \mid A = 1) \approx \frac{1}{n_t} \sum_{i:A_i=1} Y_i^{a=0} \approx \frac{1}{n_t} \sum_{i:A_i=1} \left(\underbrace{\frac{1}{|\mathcal{M}_i|} \sum_{i \text{ in } \mathcal{M}_i} Y_i}_{\text{Avg over } \mathcal{M}_i} \right)$$

Example



Example



- Conditional exchangeability holds when conditioning on Age
- For a treated individual who is 35 years old,

$$\underbrace{\mathrm{E}(Y^{a=0} \mid A = 1, \mathrm{Age} = 35)}_{\text{Can't estimate directly}} = \underbrace{\mathrm{E}(Y^{a=0} \mid A = 0, \mathrm{Age} = 35)}_{\text{Can estimate directly}}$$

Example

Job training			
Ind	Age	Y^{Train}	Y^{NoTrain}
1	20	19	?
2	25	63	?
3	38	65	?
4	38	43	?

Example

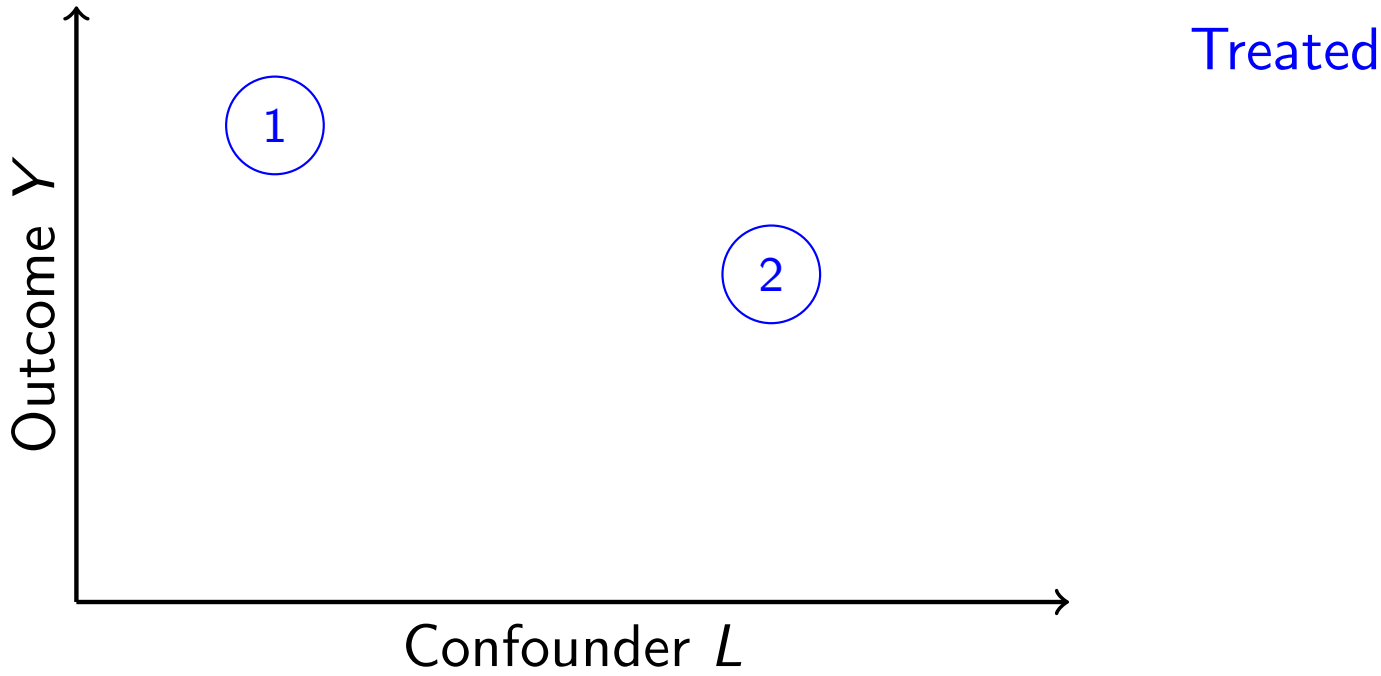
Job training

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No job training

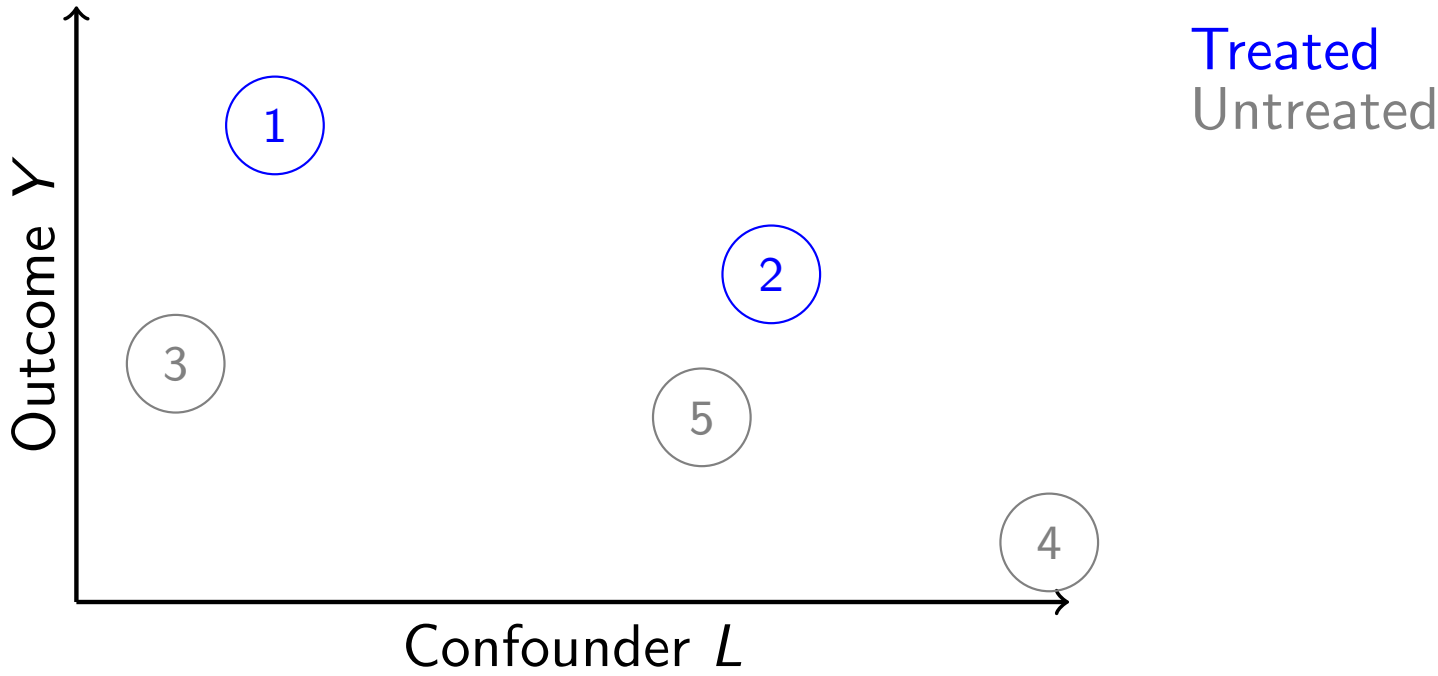
Ind	Age	Y^{NoTrain}
1	19	82
2	18	39
3	20	49
4	20	56
5	24	33
6	26	82
7	26	35
8	38	35
9	28	83
10	30	79
11	24	63
12	32	52
13	34	58
14	34	70
15	35	47
16	37	42
17	37	83
18	38	33
19	39	37
20	39	60

Matching: The big idea



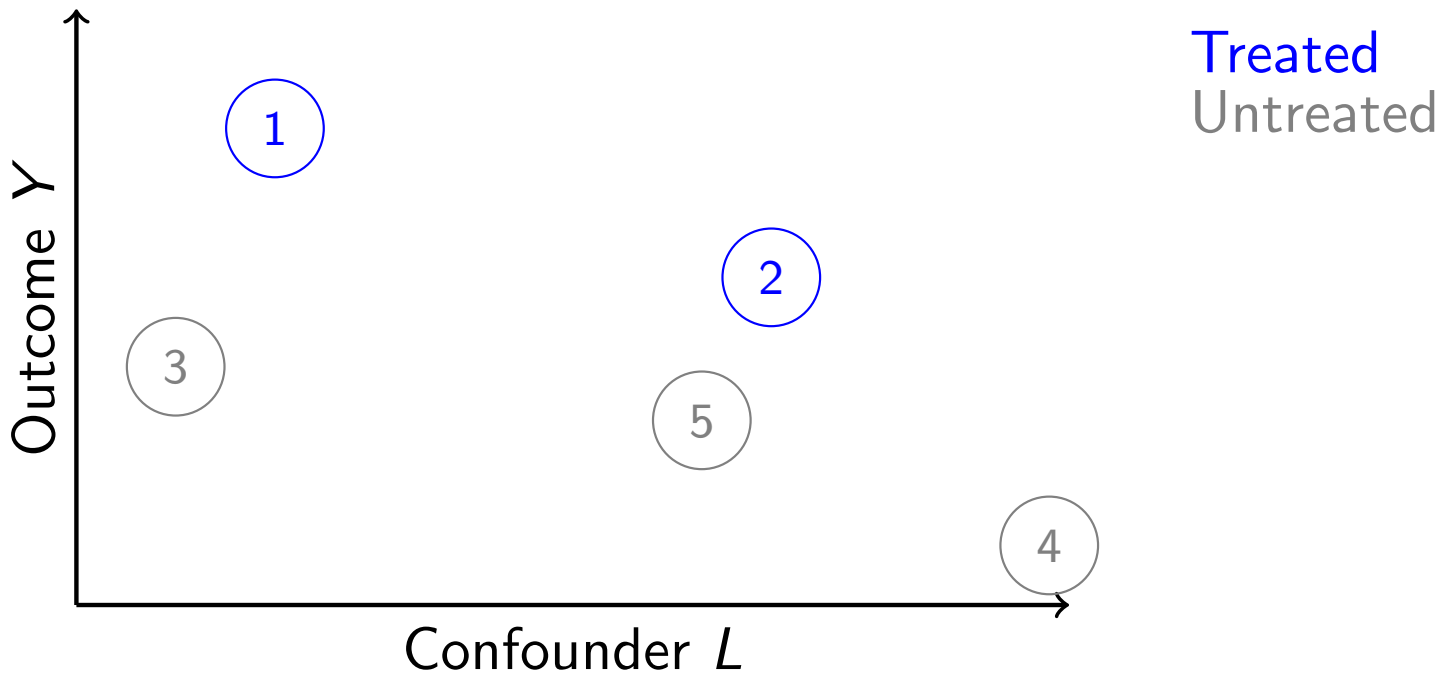
You have a some treated units.

Matching: The big idea



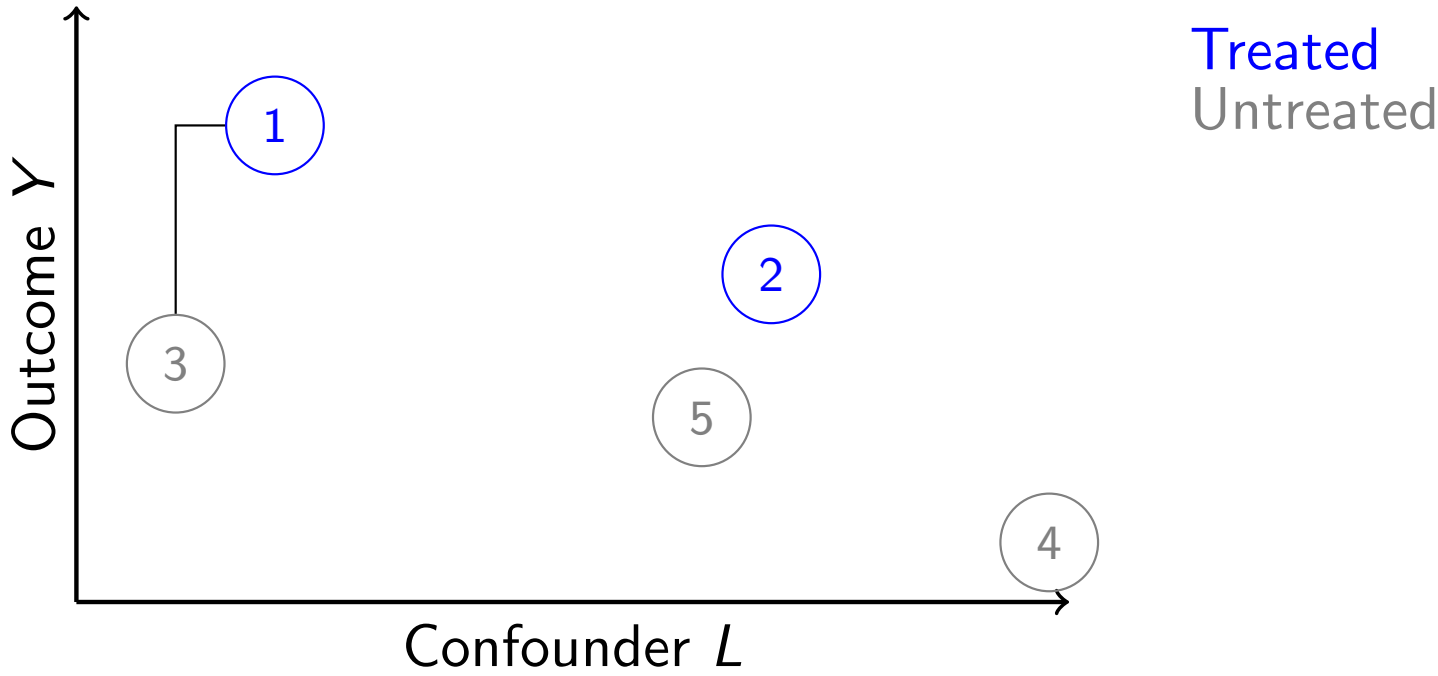
You go find some untreated units.

Matching: The big idea



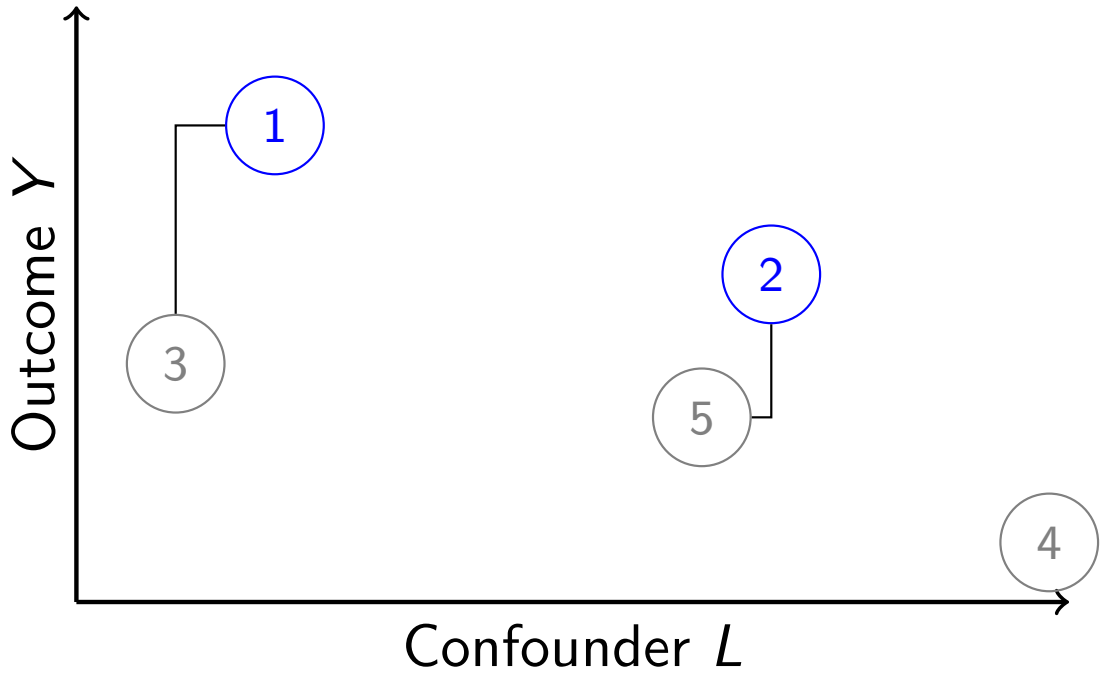
You find the closest matches along L

Matching: The big idea



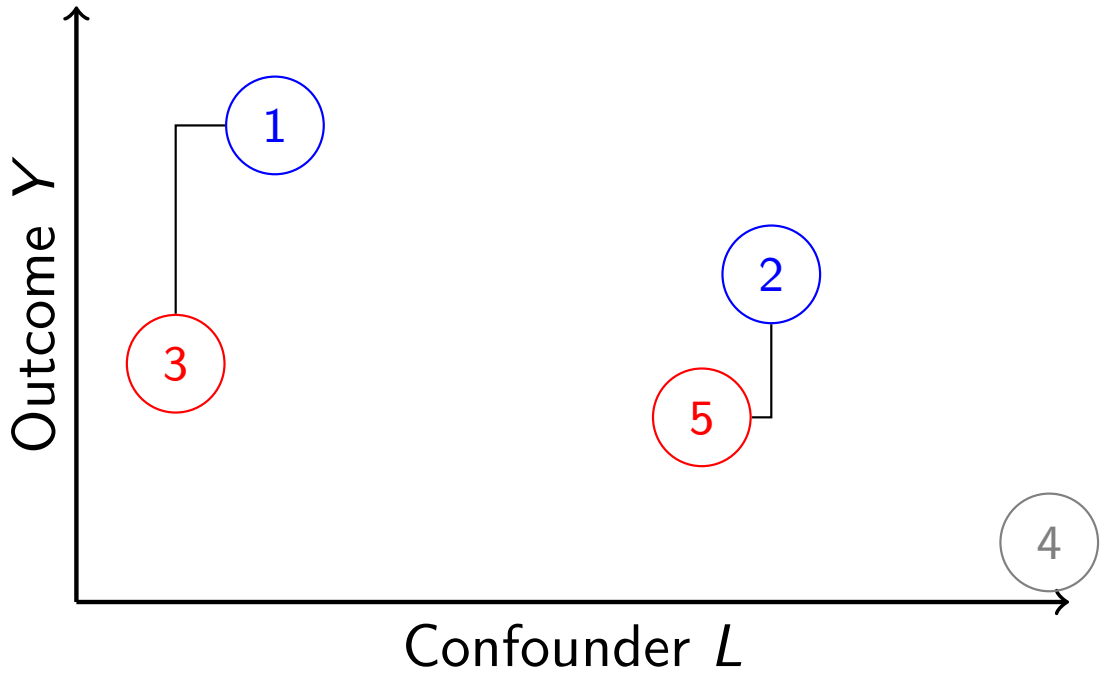
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Matching: The big idea



You find the closest matches along L

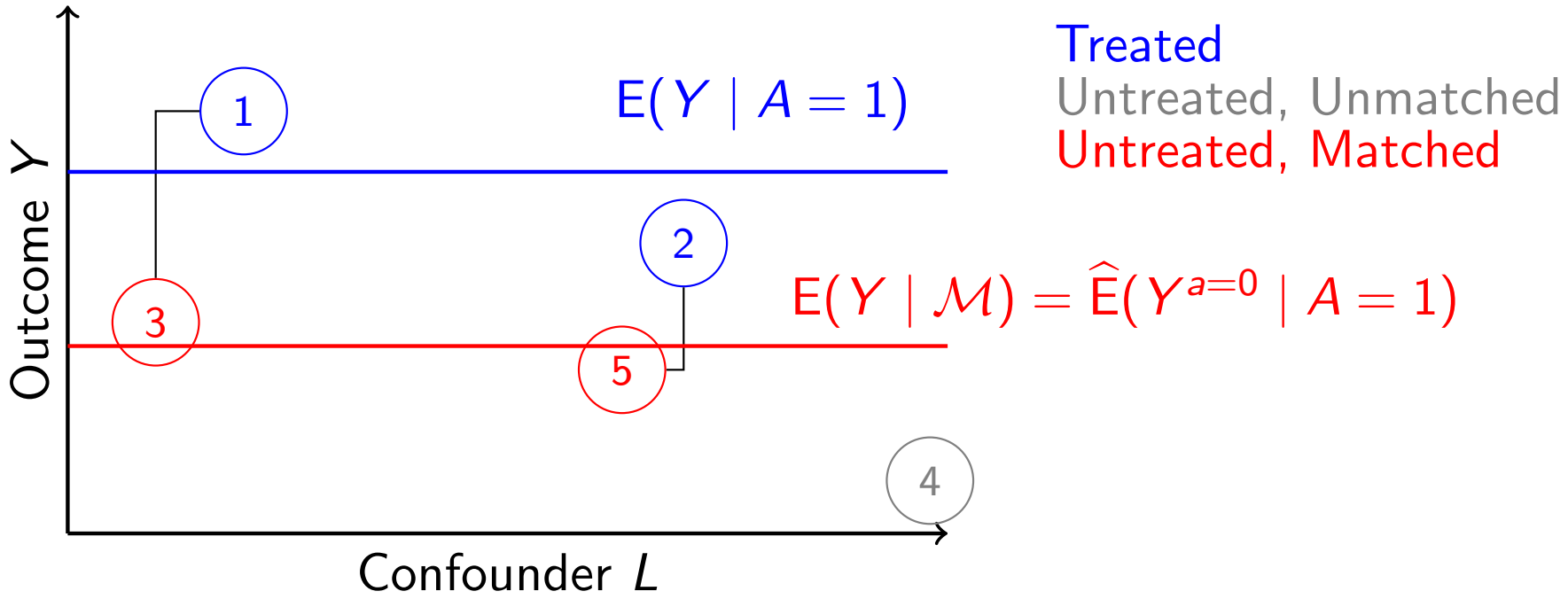
Matching: The big idea



Treated
Untreated, Unmatched
Untreated, Matched

Compare the averages

Matching: The big idea



Compare the averages

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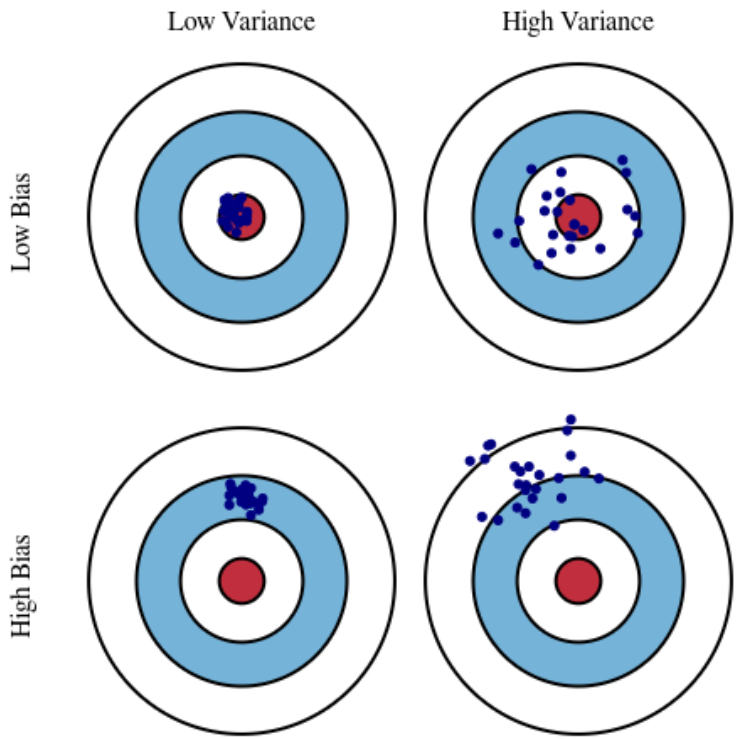
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   - We had some treated units
   - We found a set of control units which are comparable
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3. Can assess quality of matches before we look at the outcome
4. Model-free*
   - * but you have to define what makes a match "good"

Bias vs variance

The idea of matching is straightforward, but the details matter!

Bias vs variance

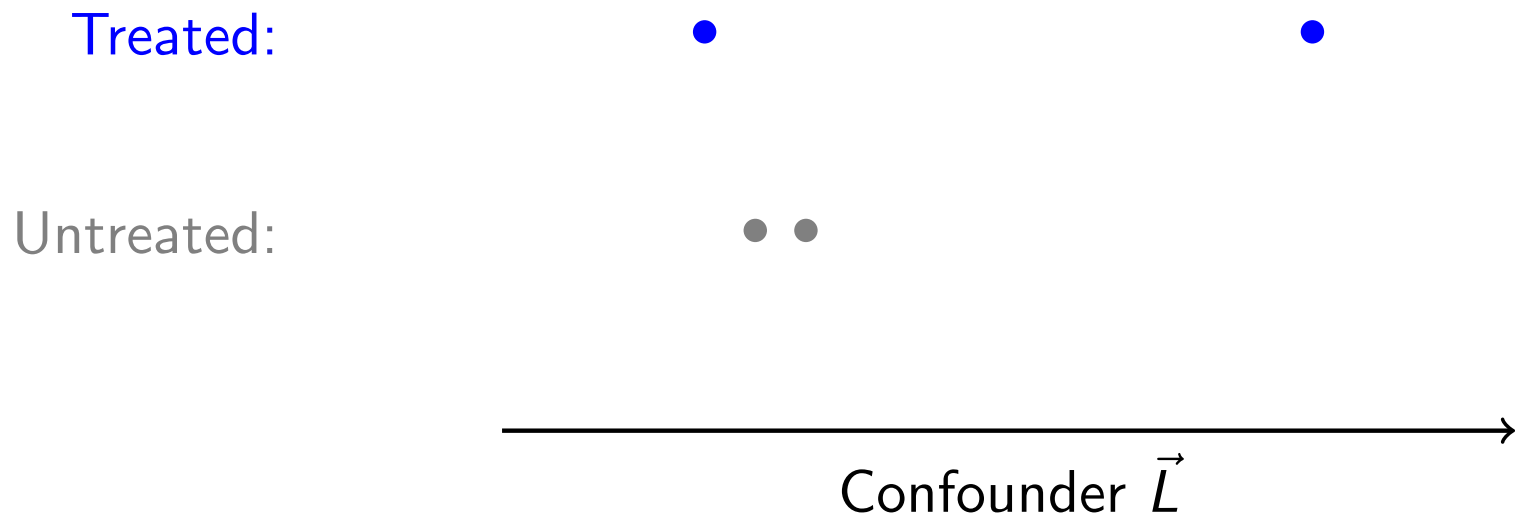
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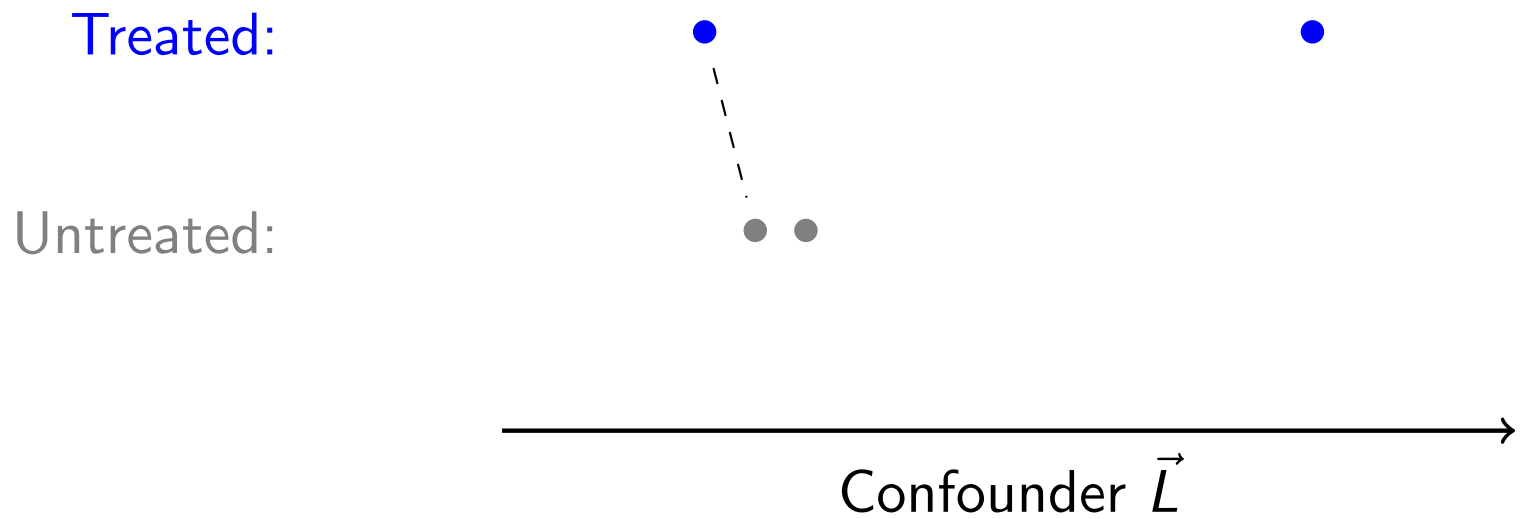
Matching in univariate settings: Algorithms

- Caliper or no caliper
- 1:1 vs k:1
- With replacement vs without replacement
- Greedy vs optimal

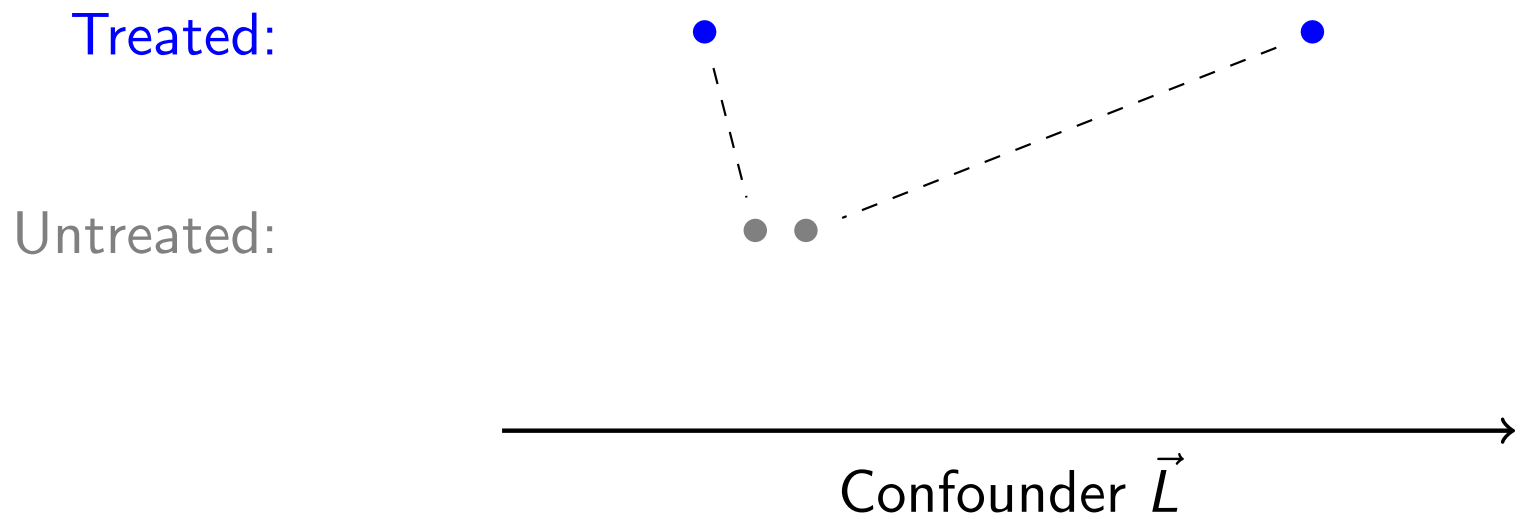
Caliper or no caliper matching



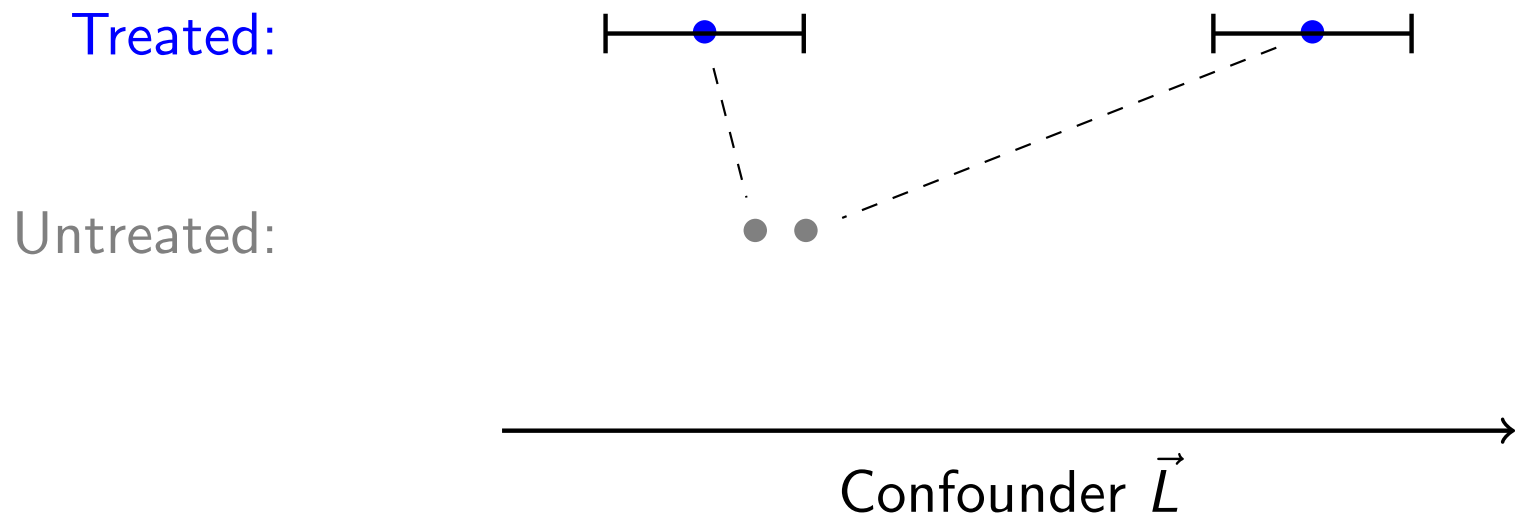
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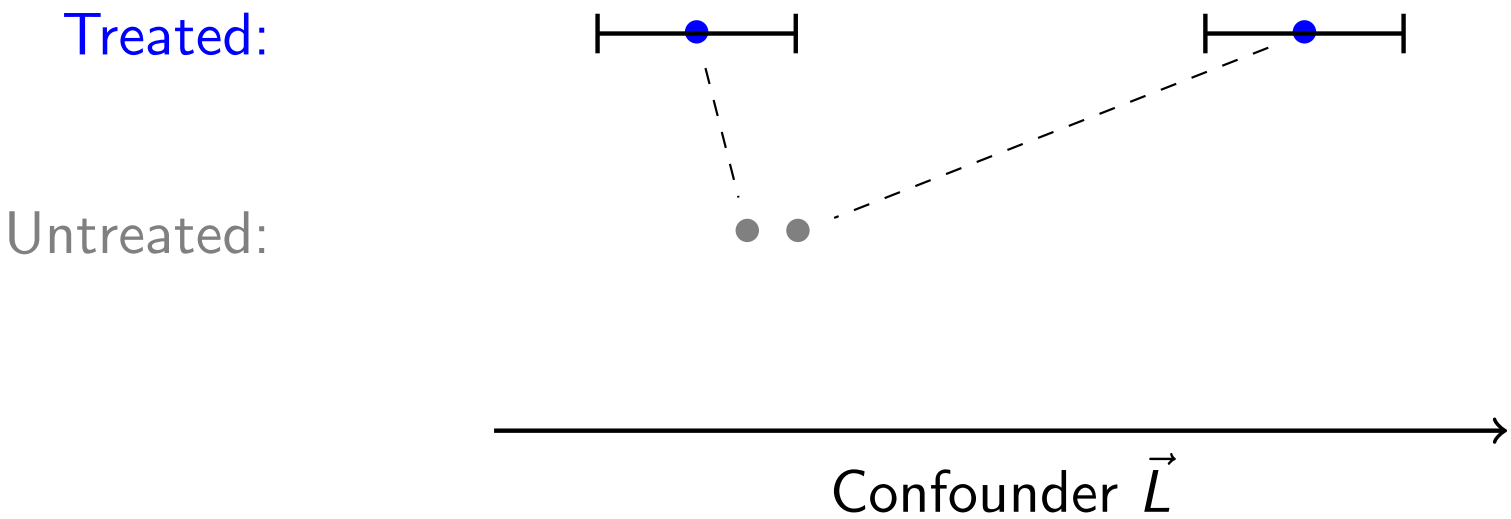
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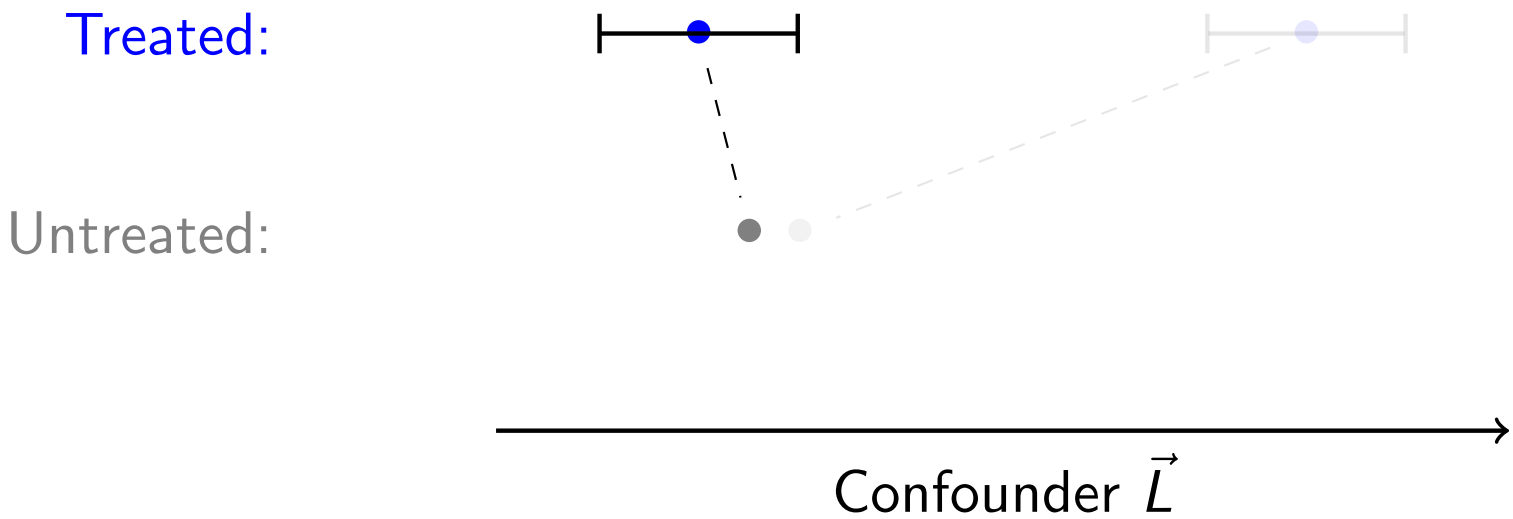


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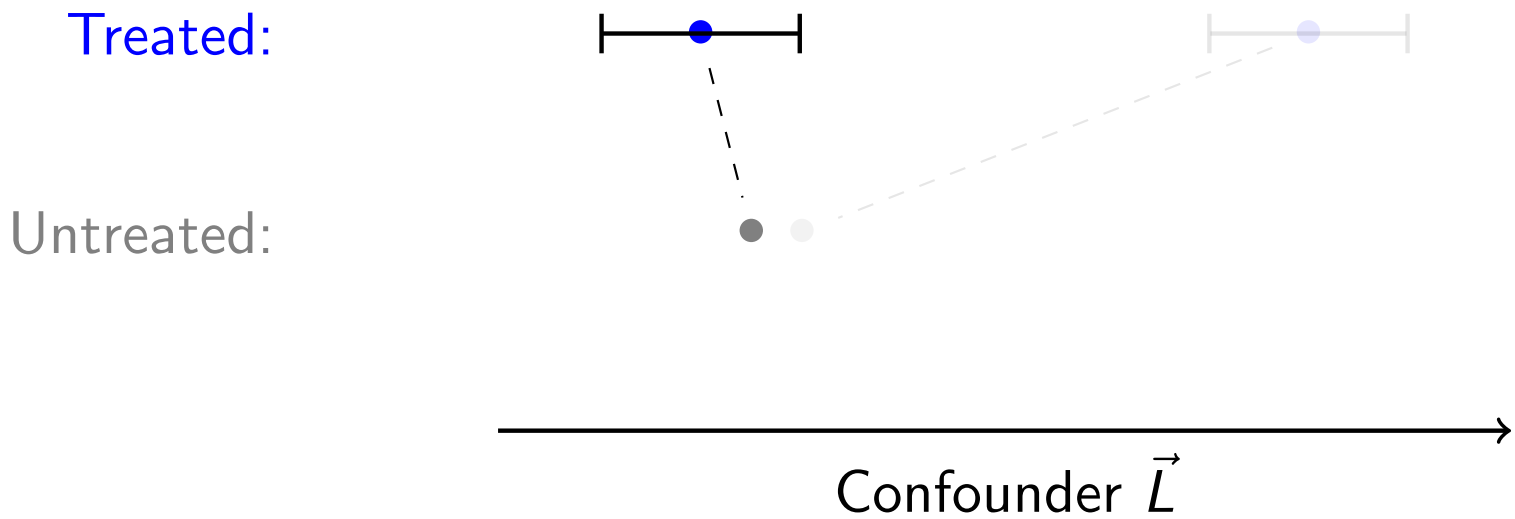
- Caliper: A radius around a treated unit such that we would rather drop the unit than make a match beyond that radius
- Poor match could be due to a positivity violation

Caliper or no caliper matching



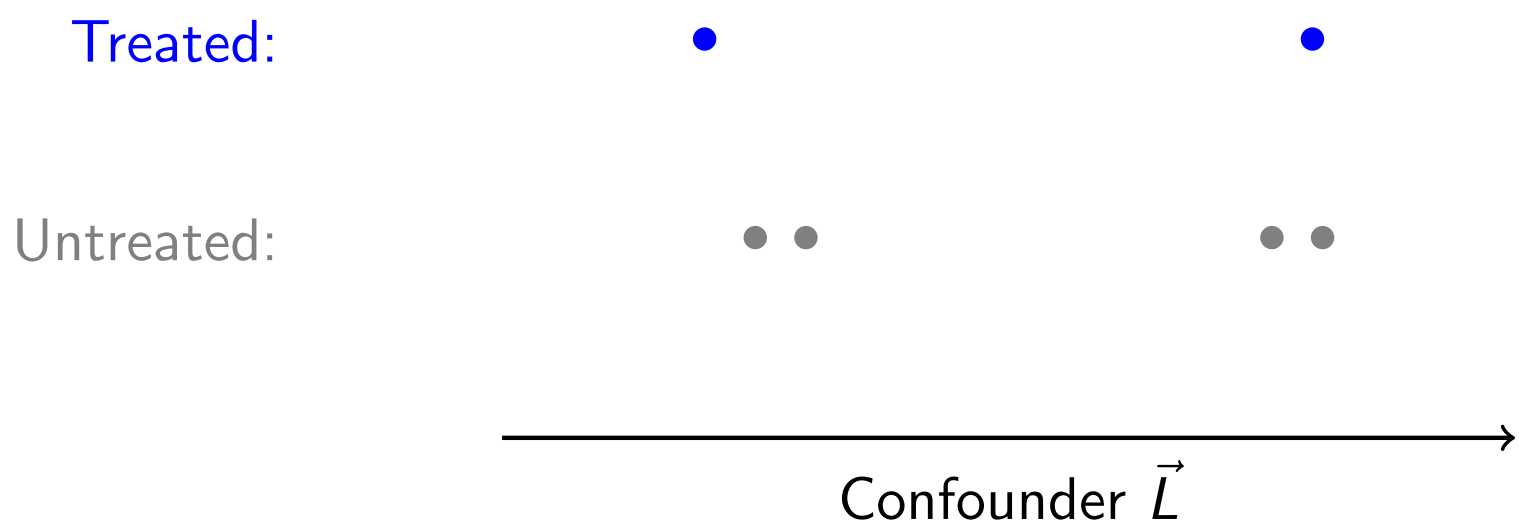
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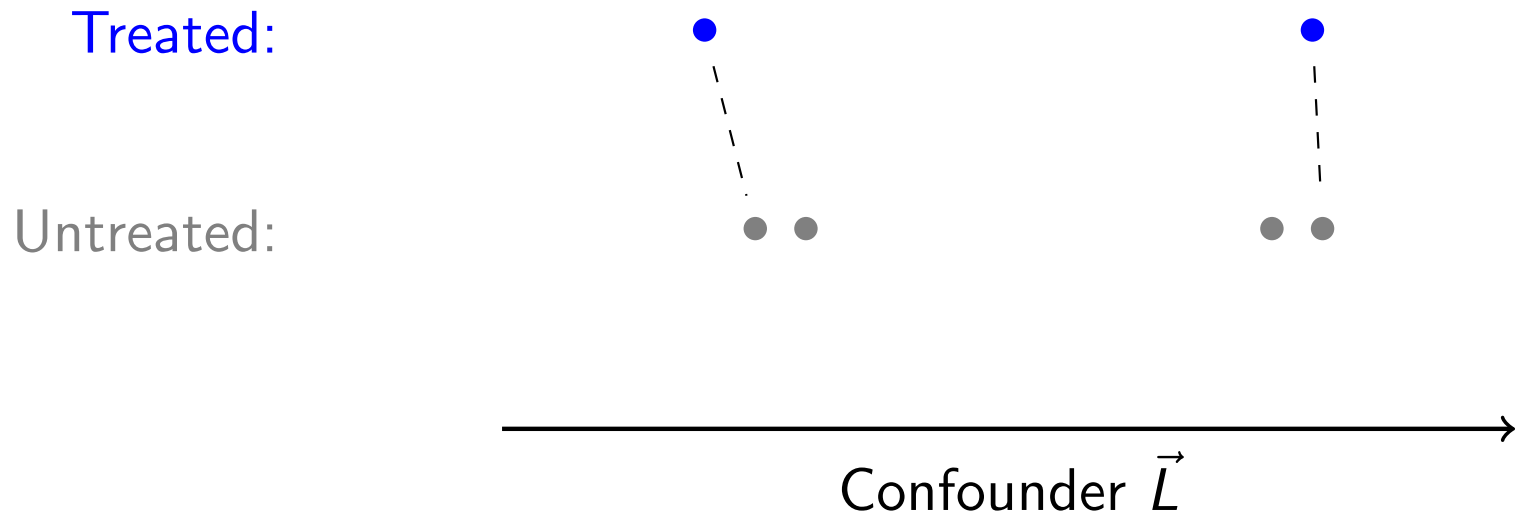


- Caliper: A radius around a treated unit such that we would rather drop the unit than make a match beyond that radius
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- Feasible Sample Average Treatment Effect on the Treated (FSATT): Average among treated units for whom an acceptable match exists

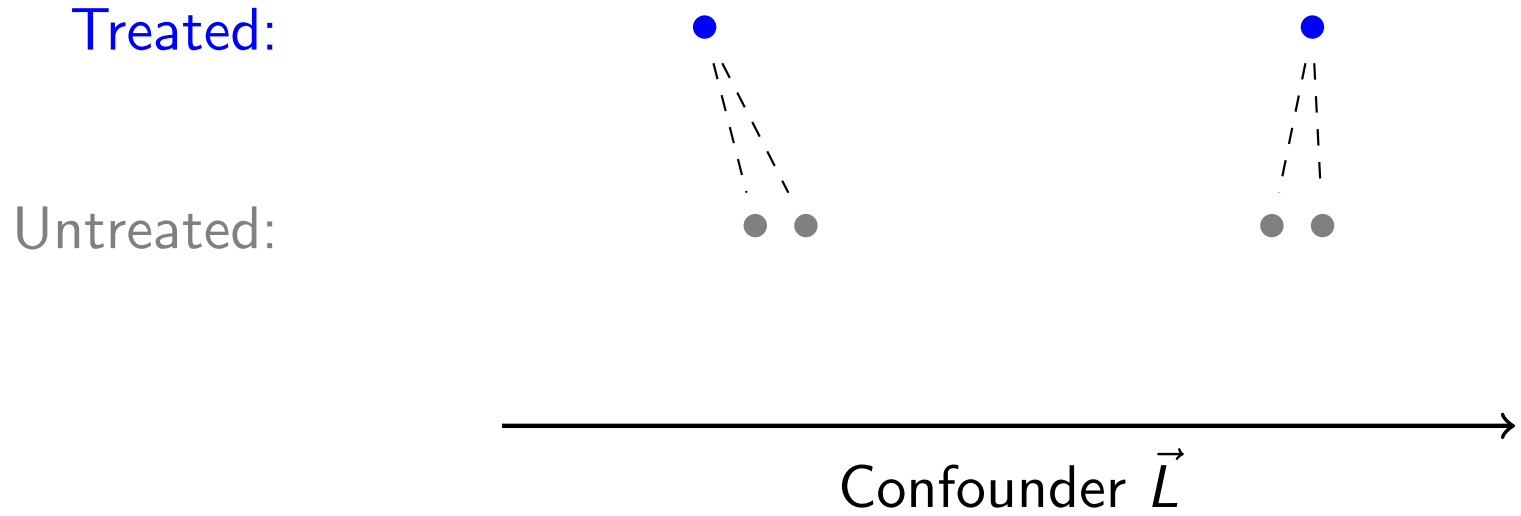
1:1 vs k:1 matching



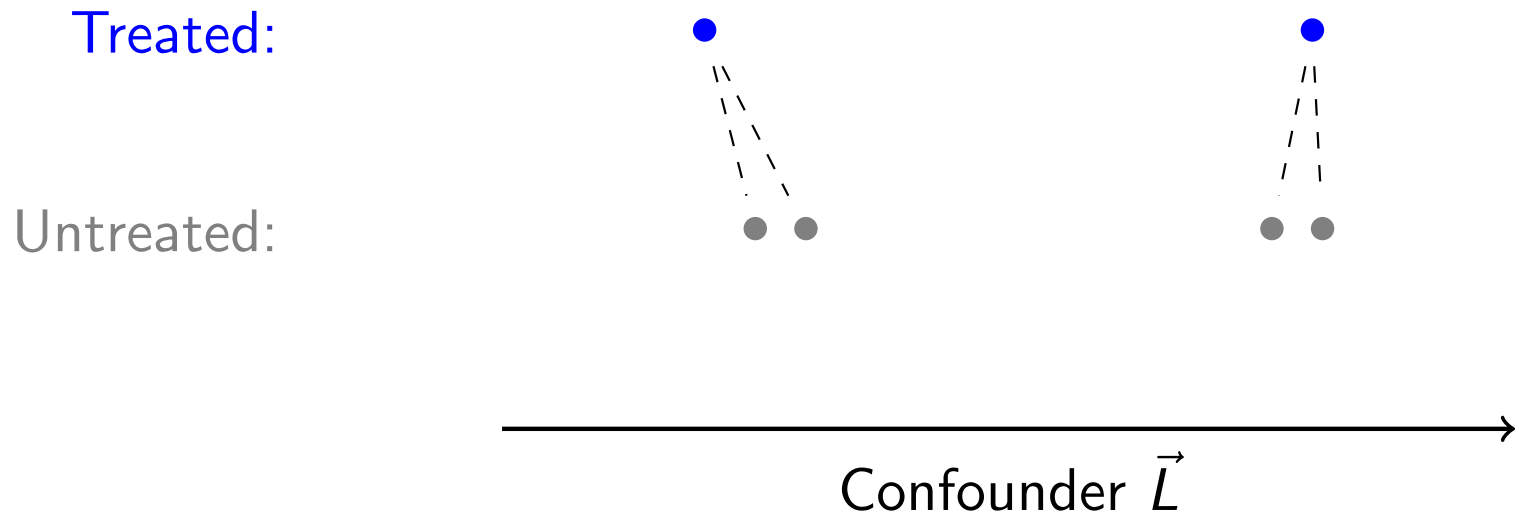
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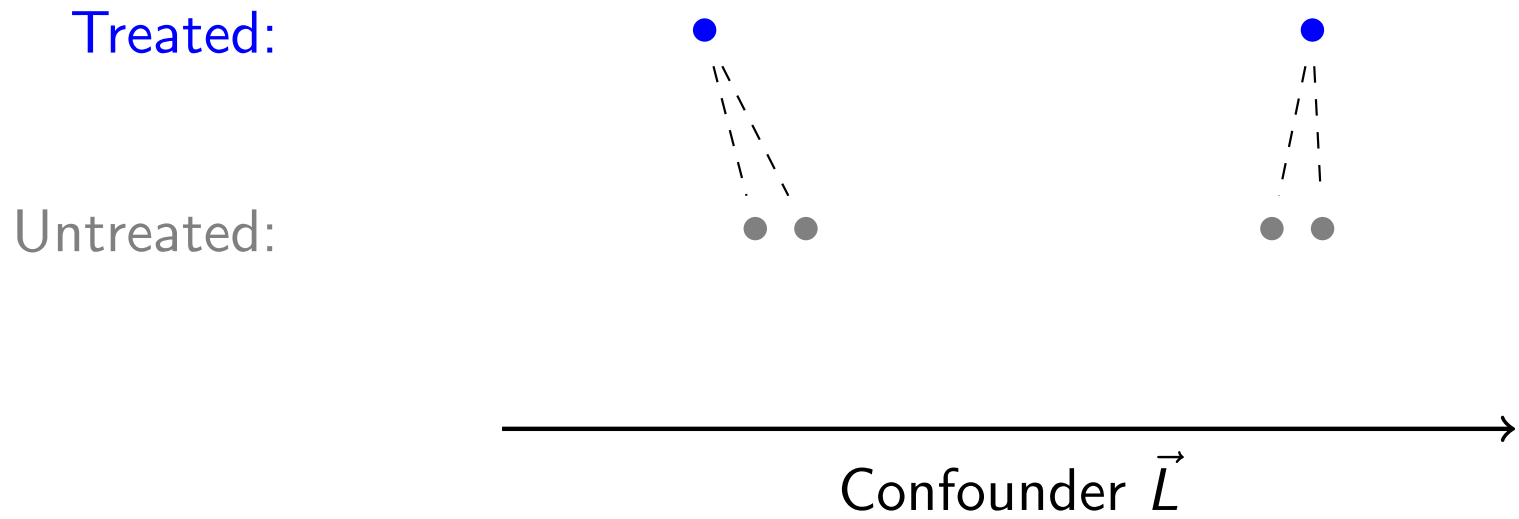


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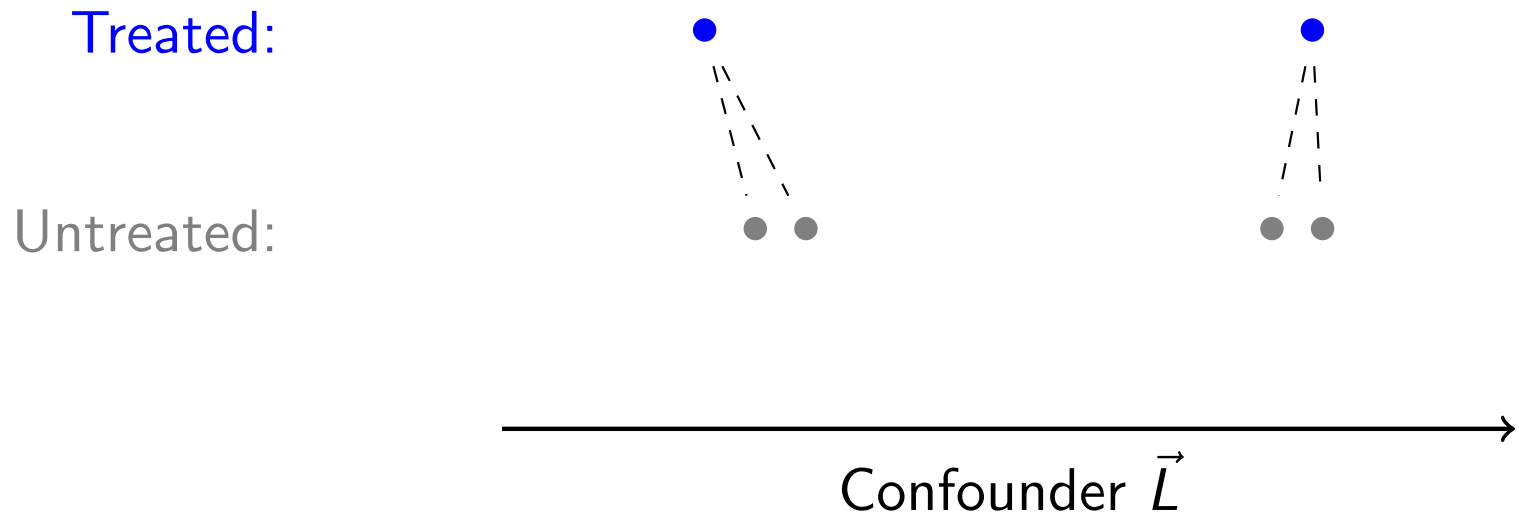
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1:1 vs k:1 matching



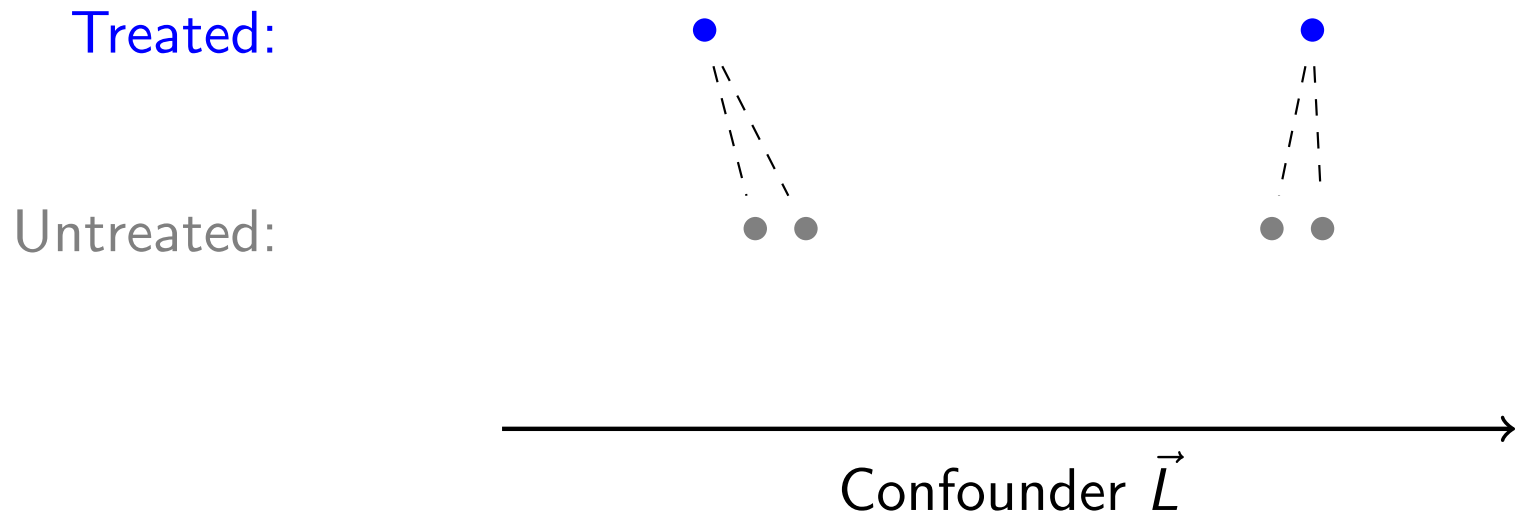
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- Benefit of 1:1 matching

1:1 vs k:1 matching



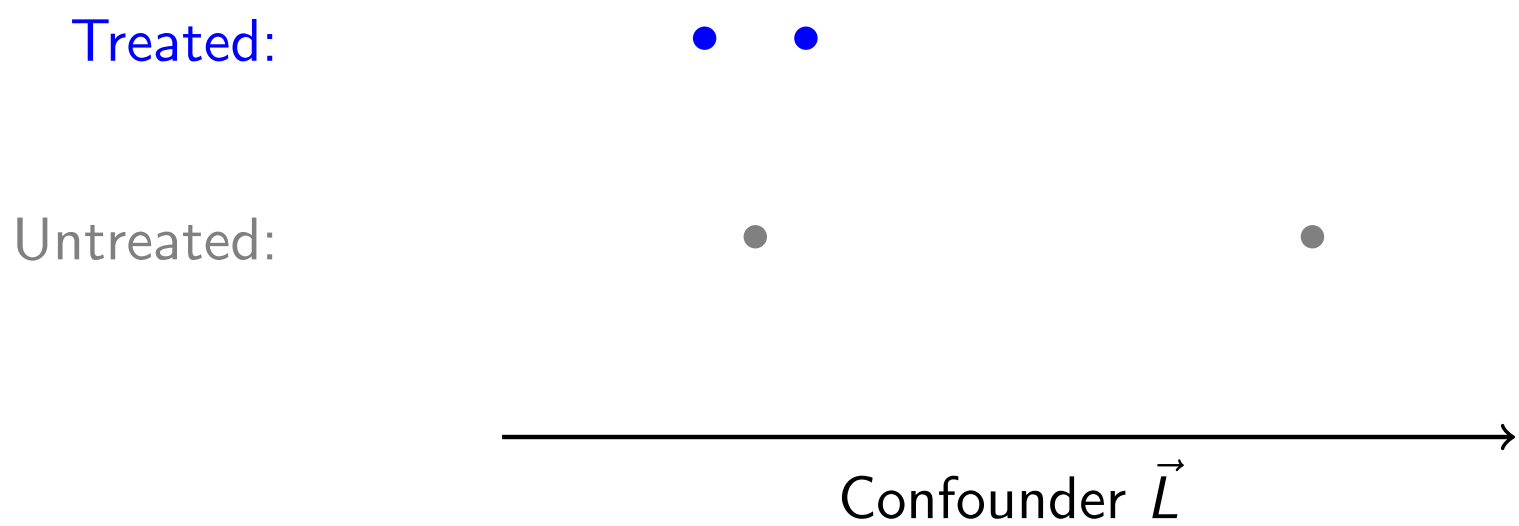
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1:1 vs k:1 matching

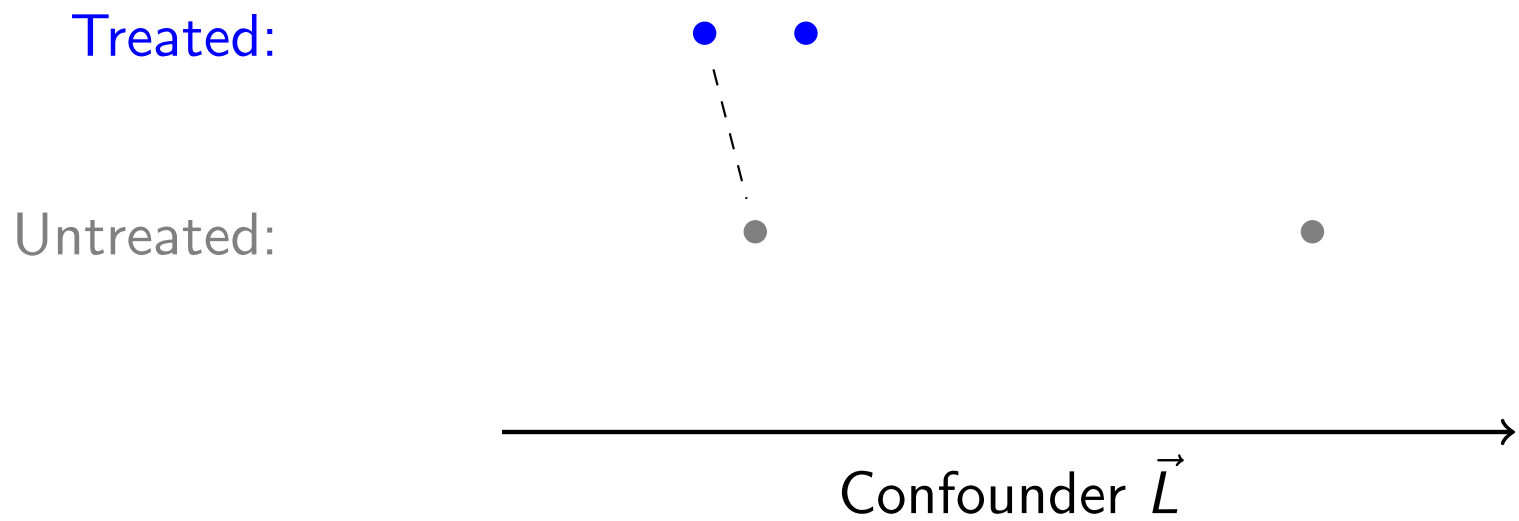


- Benefit of 2:1 matching
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- Benefit of 1:1 matching
  - Lower bias. Only the best matches.
- Greater $k \rightarrow$ lower variance, higher bias

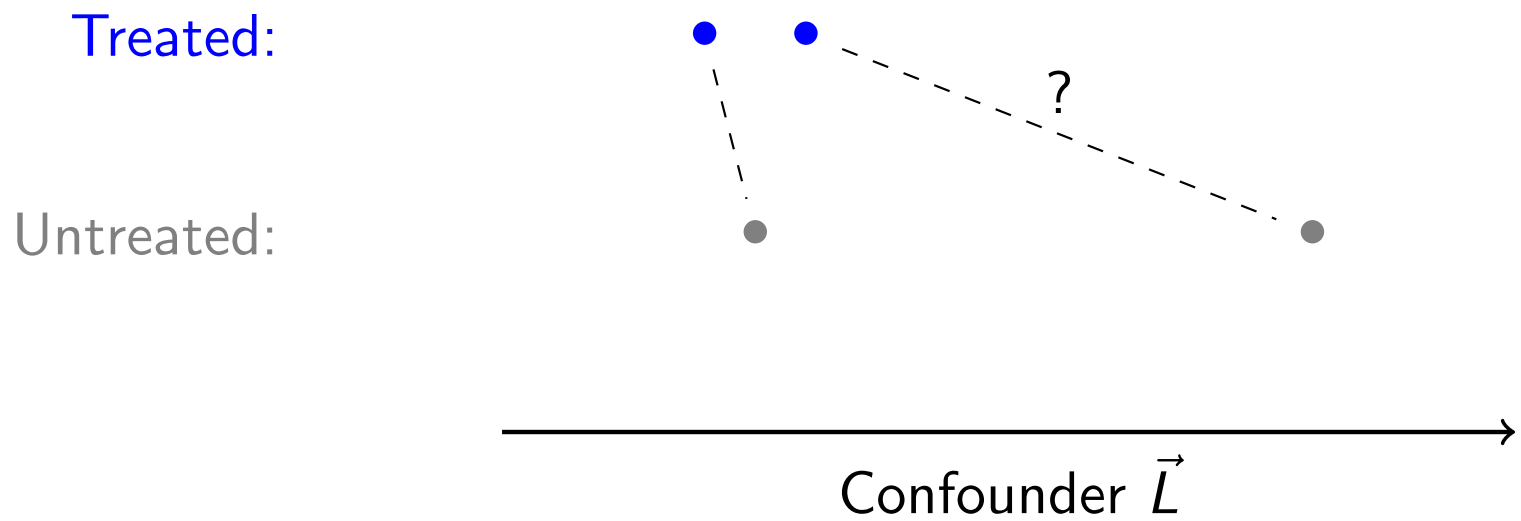
With replacement vs without replacement matching



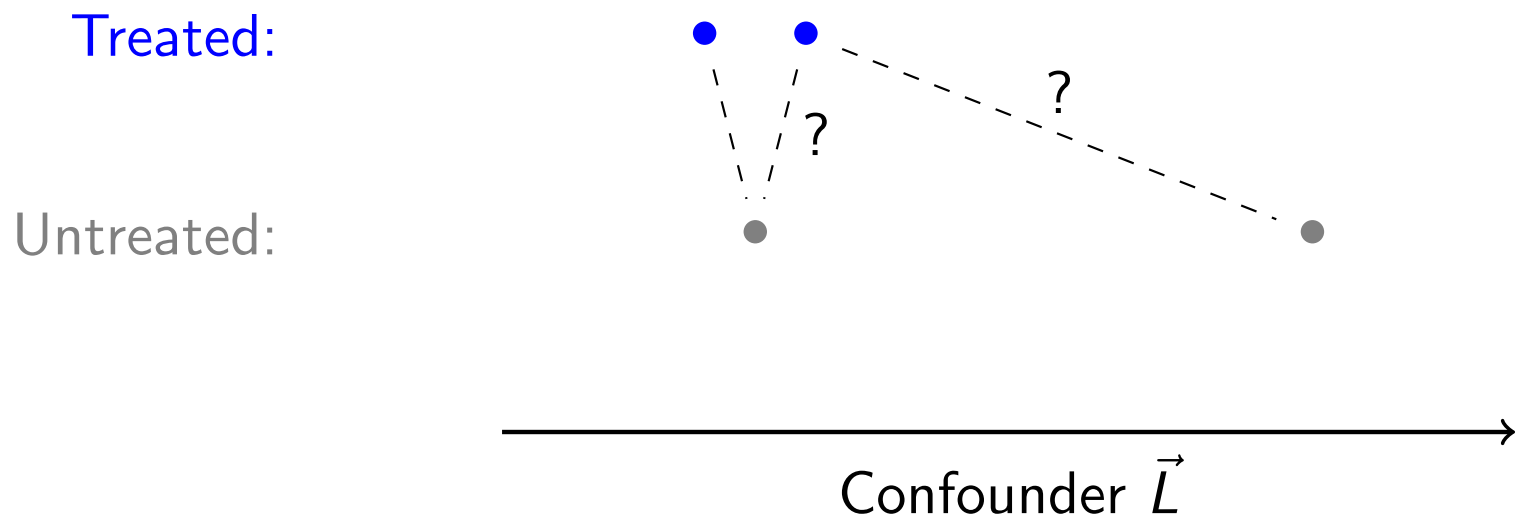
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With replacement vs without replacement matching

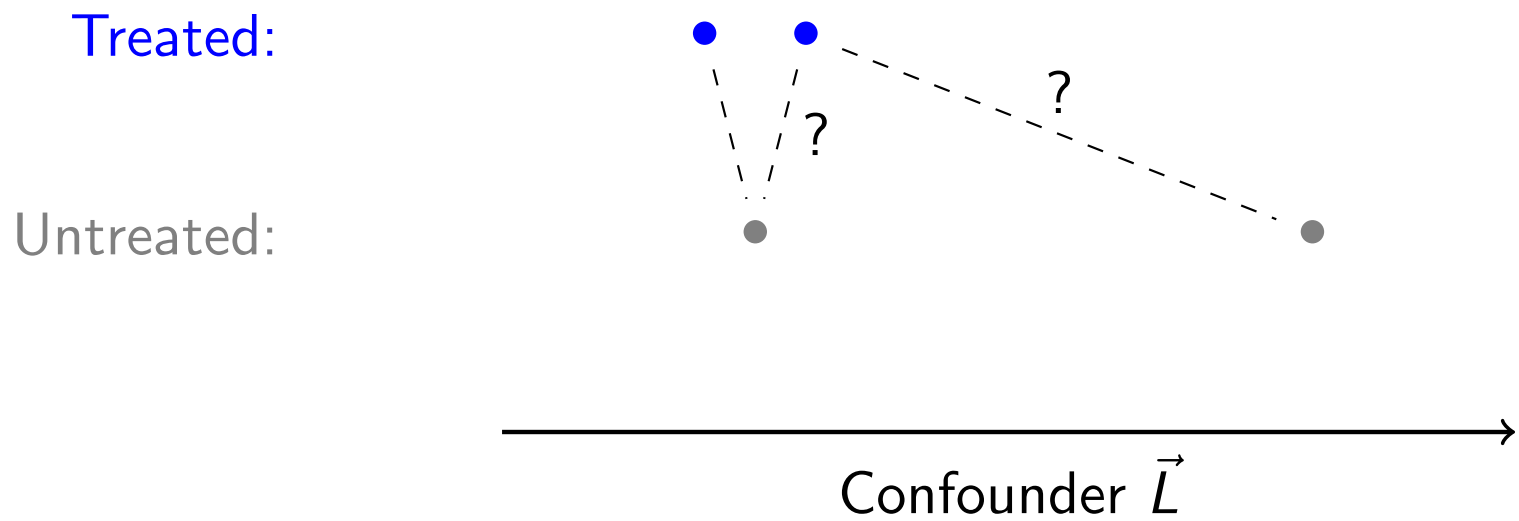


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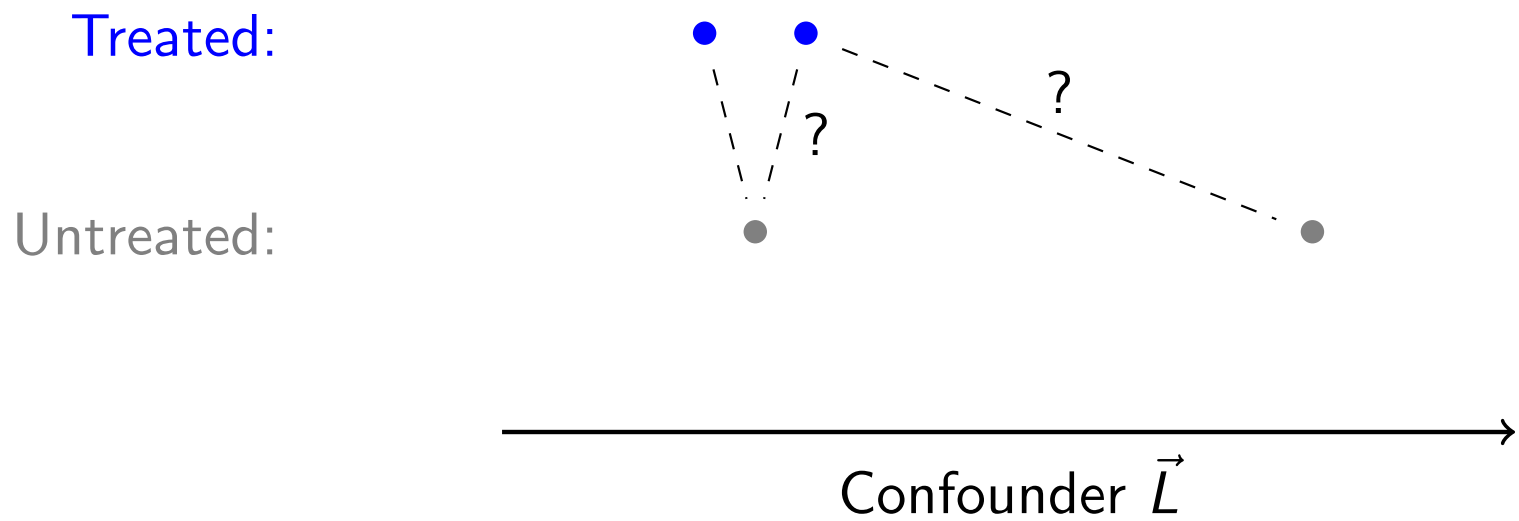
- Benefit of matching without replacement
- Benefit of matching with replacement

With replacement vs without replacement matching



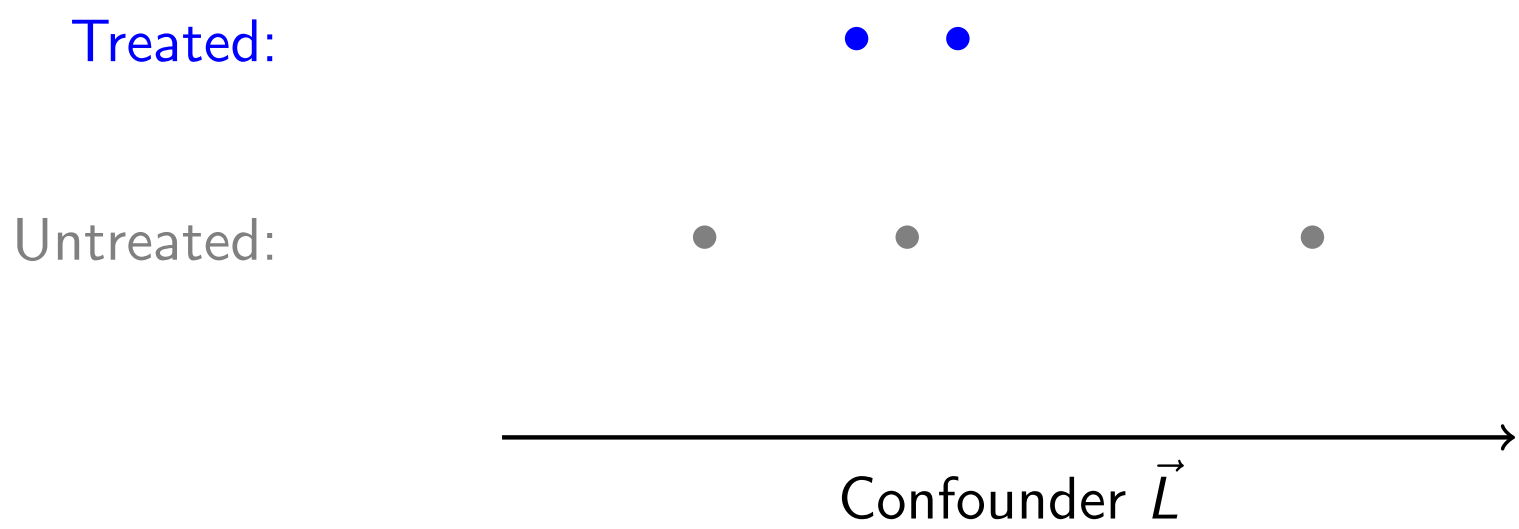
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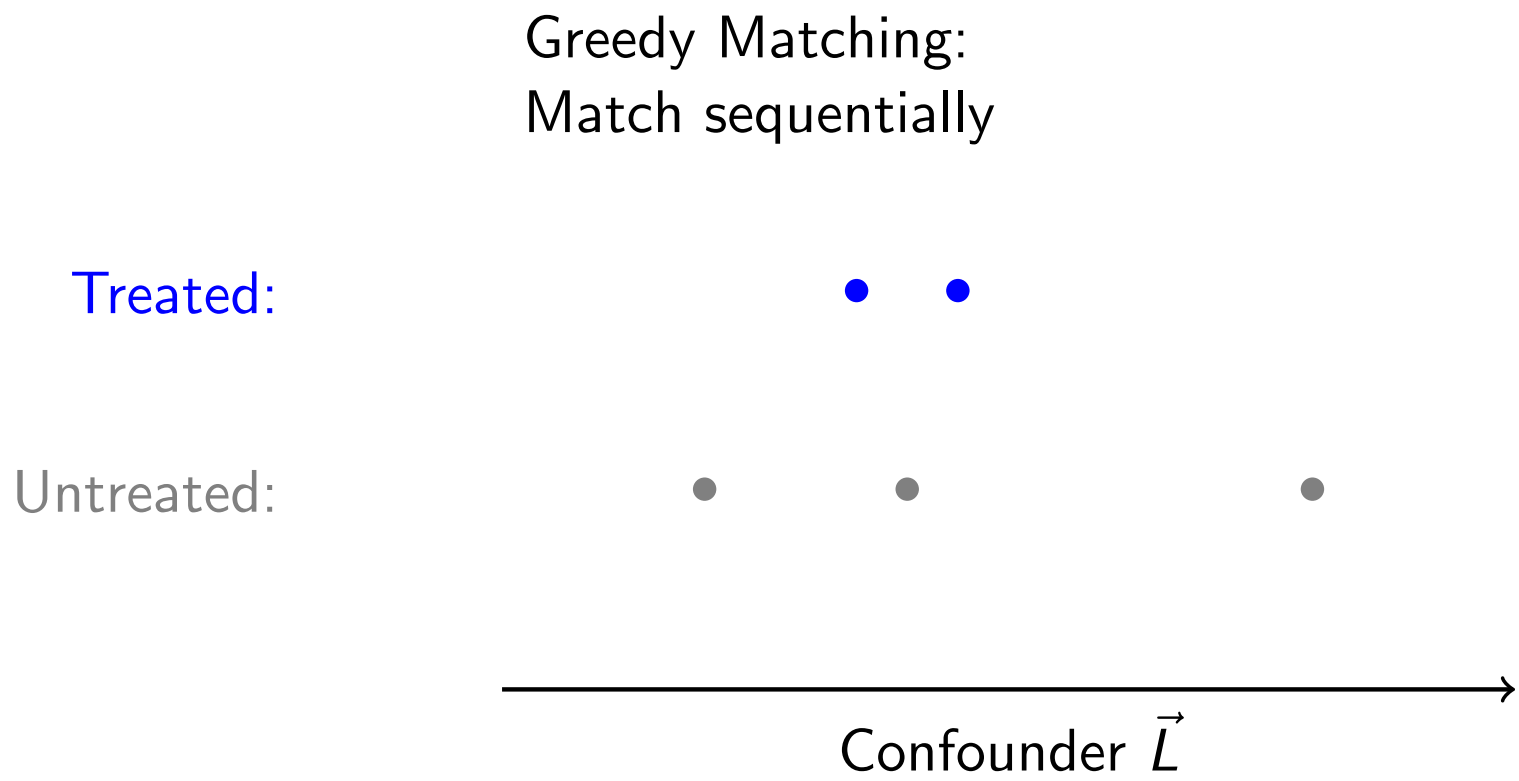
- Benefit of matching without replacement
  - Lower variance. Averaging over more cases.
- Benefit of matching with replacement
  - Lower bias. Better matches.

Greedy vs optimal matching[1]



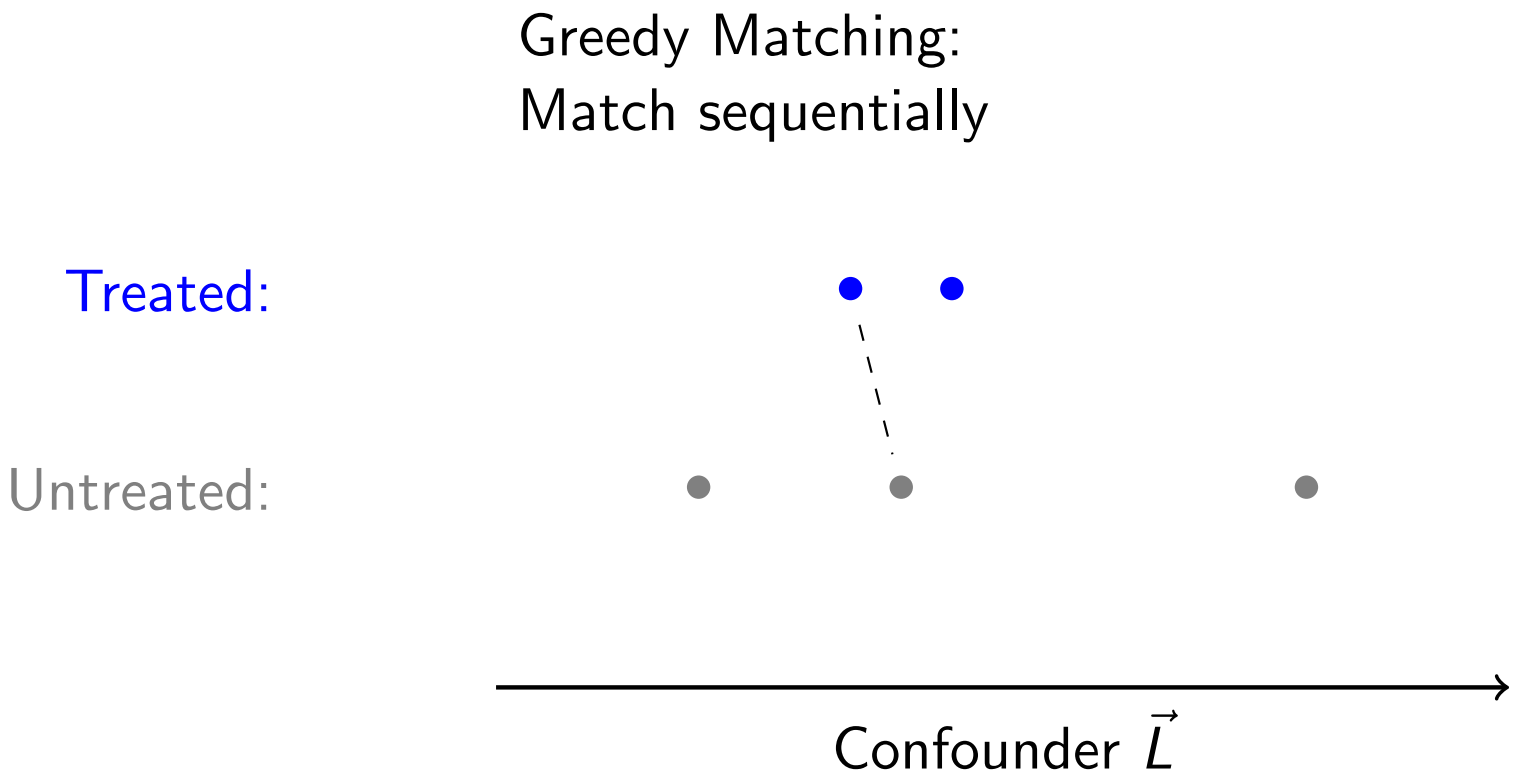
[1] Gu, X. S., & Rosenbaum, P. R. (1993). Comparison of multivariate matching methods: Structures, distances, and algorithms. Journal of Computational and Graphical Statistics, 2(4), 405-420.

Greedy vs optimal matching[1]



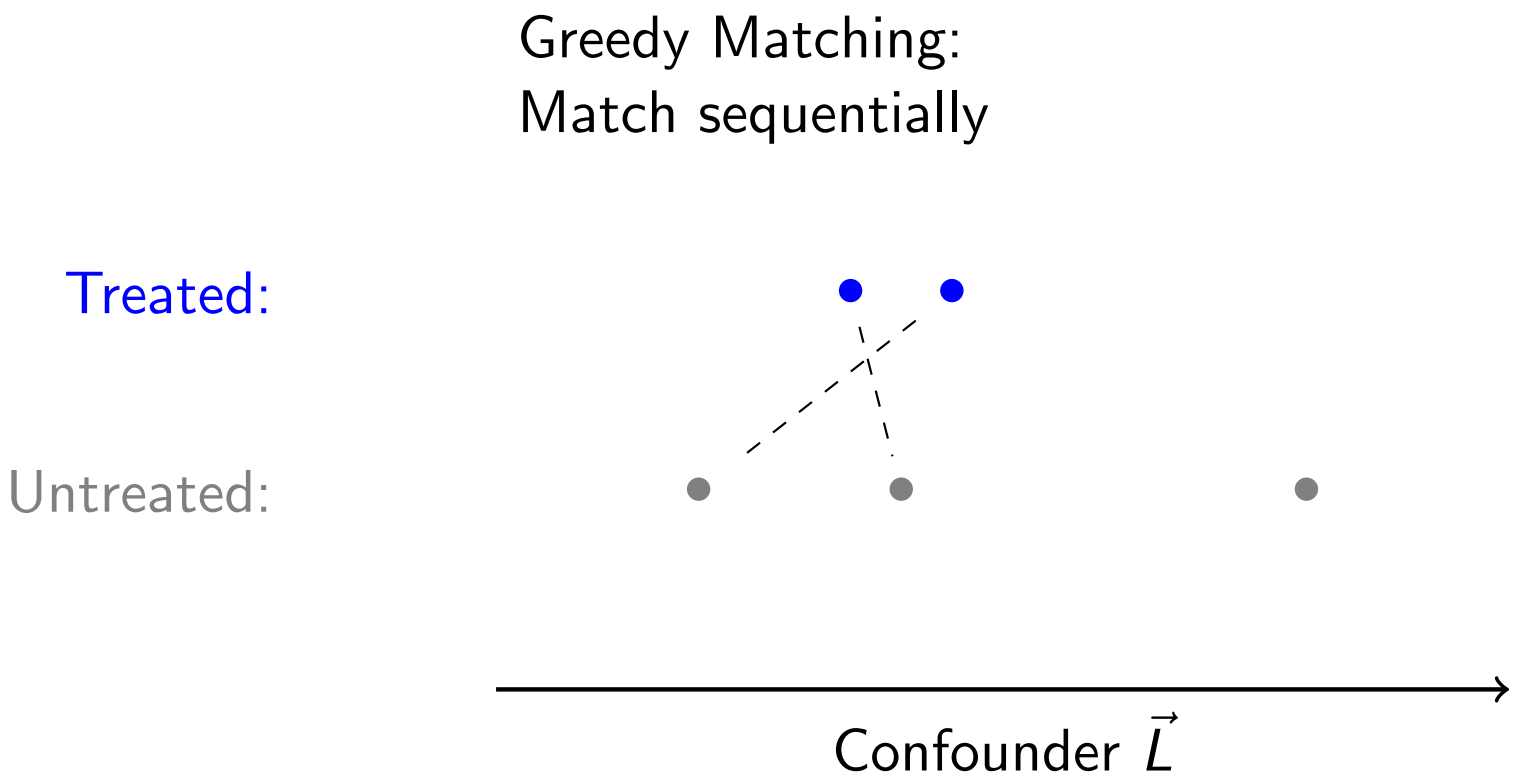
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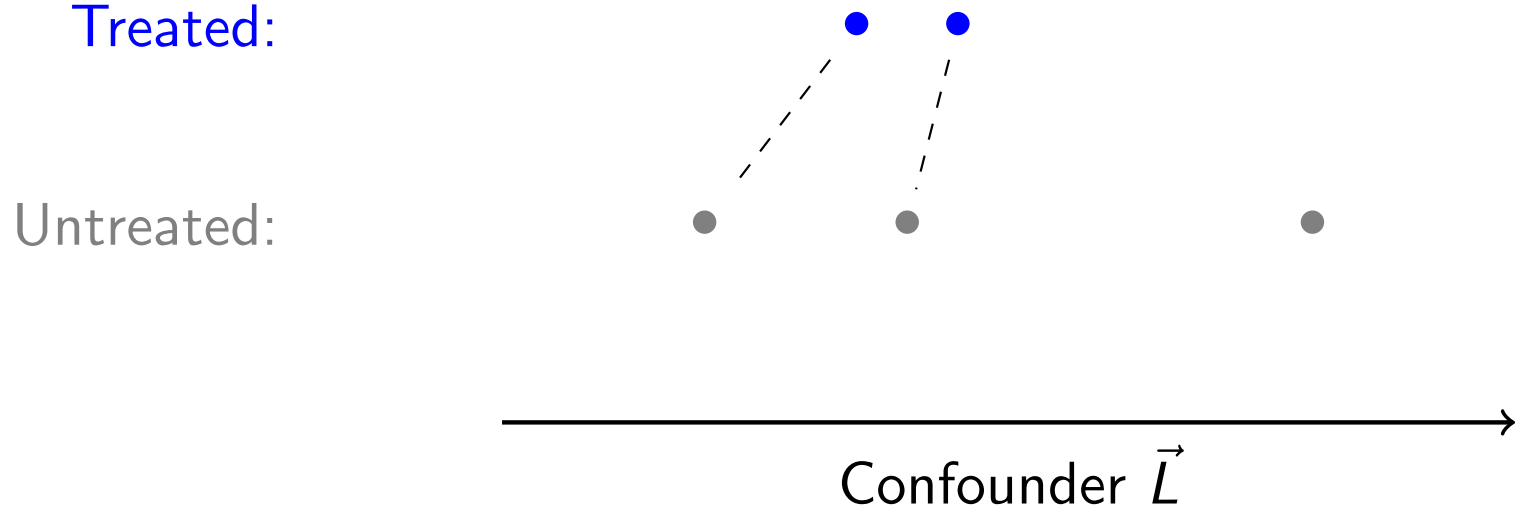
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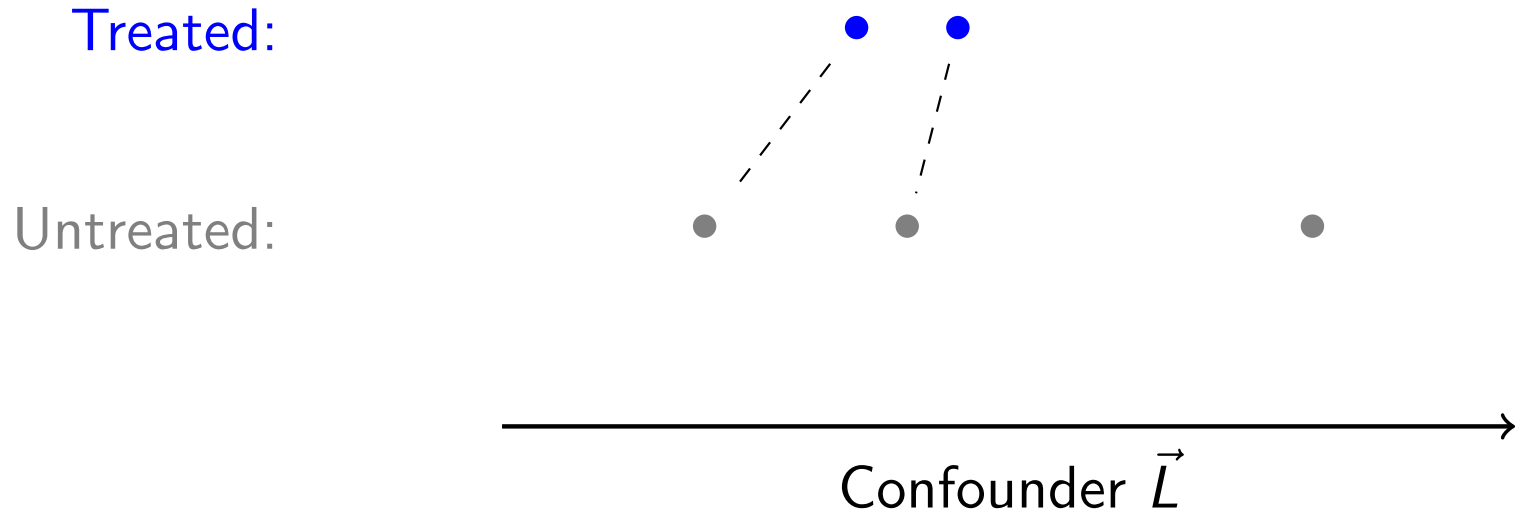
Optimal Matching:
Consider the whole set of matches



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Consider the whole set of matches



- Optimal is better. Just computationally harder.

[1] Gu, X. S., & Rosenbaum, P. R. (1993). Comparison of multivariate matching methods: Structures, distances, and algorithms. Journal of Computational and Graphical Statistics, 2(4), 405-420.

Matching in univariate settings: Algorithms

- Caliper or no caliper
- 1:1 vs k:1
- With replacement vs without replacement
- Greedy vs optimal

Matching in univariate settings: Algorithms

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Many reasonable choices, good choices depend on the data you have

Learning goals for today

At the end of class, you will be able to:

1. Explain how matching can be used to estimate causal effects
2. Explain bias variance trade-off in various matching procedures